Executive Summary

Dated 02/23/2022

Introduction

The purpose of this summary is to document the parallel testing that was conducted between Actimize Xceed Evidence Lake (Actimize) and Hummingbird Case Management (HCM). Additionally, the details of the analysis related to the Actimize rules that was conducted and the pre-implementation tuned that occurred in accordance with ABC Bank's (ABC) risk tolerance and in line with capturing suspicious activity that was identified in HCM will be included in this report. As part of the Actimize transaction monitoring implementation, rule segmentation was also analyzed.

A risk-based approach was used to analyze the efficiency of current settings in Actimize and based on the results, recommendations were made to improve the alert quality. In line with ABC's tuning methodology, which focuses on reducing false positives and conducting Below the Line (BTL) testing, pre-implementation tuning exercises were completed. The tuning exercises were conducted using processes outlined in the Model Tuning and Change Control Procedure.

Scope/Tuning Approach and Objective of the Parallel Testing

Prior to starting the parallel testing, all available rules in Actimize were reviewed to determine which rules were comparable to the rules currently being used to generate alerts in HCM. Based on this review, 16 rules were selected to be included in the parallel testing.

The parallel testing included all active rules in Actimize and HCM. During the pre-implementation tuning of rules in Actimize, all active rules for ABC and ABCX were evaluated. Additionally, an evaluation of inactive rules was completed to determine if any should be activated. The tuning approach that was used during this exercise was to analyze triaged alerts in Actimize and HCM, completed cases and SARs to measure the performance of the rules.

Parallel Objective:
Parallel outcomes analysis, under which both the HCM and Actimize are tested against realized outcomes, will be used to determine appropriate adjustments, if any, to the Actimize model. If adjustments are made, the models will be tested and analyzed again.
Analyzing Steps:

1) Analyze the money laundering/terrorist financing (ML/TF) SARs that were generated from HCM alerts and determine if an Actimize alert was also received on the customer.
   - Purpose: to determine if Actimize is capturing all suspicious transactional activity related to money laundering/terrorist financing (ML/TF) that warranted a SAR filing.
2) Analyze Actimize alerts to determine if a HCM alert was also generated.
   - Purpose: is to review the overlap between models and to assist with rule tuning.
3) For Actimize alerts that didn't have HCM alerts, review these alerts for quality
   - Purpose: to gauge the effectiveness of the Actimize rules.

Parallel Scope of Testing:
Cycle 1 – Oct 1 to Nov 30, 2021
Cycle 2 – Dec 2021
Cycle 3 – Jan 2022

Supporting Documents

The following documents were used to assist with the evaluation of the various ABC and ABCX customer base/product and services, transaction types and limits and to analyze the current thresholds and parameters:

- ABCX Program Matrix -v01.18.2022
- Xceed Evidence Lake Reference Manual 4.28.2021
- Xceed Evidence Lake Entity Group Addendum 2.23.2022
- Xceed Evidence Lake Reference Manual 2.24.2022
- Xceed_Evidence_Lake_Data_Dictionary_v3.0
- AML Rule Typology V1
- Hummingbird Case Management System:
    - AML Rule Set Directory 10.01.2021
    - RuleSet_Prod _05.03.2021
    - RuleSet_Prod _Aspiration_05.03.2021
    - RuleSet_Prod_ABC_Business_05.03.2021
    - RuleSet_Prod_OneFinance_06.28.2021
- Transection Mapping V6 to V12
- ABC Rules for Segmentation - 02.14.2022
- Reports used for Actimize Analysis
    - Actimize Alerts- Sept. 2021
    - Actimize Alerts- Oct.-Nov. 2021
    - Actimize Alerts-Dec. 2021
    - Actimize Alerts-Jan. 2022
    - Case Determinations Dec 2021-updated 2.14.22
    - Parallel Testing Alerts-Cases Jan & Feb 2022
    - Cases ABC Jan-Feb 2022
- Parallel Summary Analysis
- Current Rules and Conditions - Recommendation dated 01.21.2022

System Rule Comparison

The rules in Actimize are comprised of 16 rules. The rules consist of one or more conditions. A condition has several configurable parameters such as transaction type, dollar amount threshold and look back period. All transactions are screened against each condition. Transaction monitoring alerts are generated as a result of executing the transaction monitoring rules and when transactions meet the rule's conditions. Similarly, rules in HCM are made up of 30 rules and are used to screen transactions. Both systems are built on AML typologies that include monitoring for the following activity:

- Out of pattern
- Excessive
- Structuring
- Velocity
- Fraud
- Keyword

Summary of Actimize Rules:

Actimize	AML Typology Category
Inactive Account Fraud (1 condition)	Fraud
New Account Fraud (4 conditions)	Fraud
Transaction Type with Cash (1 condition)	Velocity/Structuring
Historical Behavior Comparison	Out of Pattern
Elder Abuse	Fraud
Overpayment of Loans Aggregated	Excessive
Repayment of Loan Soon After Loan Release	Velocity
Repayment of Loan within Percentage of Loan Period	Velocity
Dormant Account Aggregate	Fraud
Structuring	Structuring
Dormant Account Single	Fraud
Activity Deviation	Out of Pattern/Velocity
Risky International Transfer	Keyword
Detect Excessive Transaction Activity	Excessive
Velocity Aggregate	Velocity
Burst in Beneficiary Activity	Out of Pattern

AML Risk	Working Standard	Hummingbird System	Actimize System	Scenario
Transaction Laundering and Layering, Pass-through, Collecting payments for illicit activities	Monitor for out of pattern transactions	ACH Out of Pattern, ATM Out of Pattern, Cash Out of Pattern (Cash tag), Cash Out of Pattern, Check Out of Pattern, Monetary Instrument Out of Pattern, POS Out of Pattern, Wire Out of Pattern	- Activity Deviation - Burst in Beneficiary Activity - Elder Abuse - Historical Behavior Comparison	- Identify significant change in a customer Incoming or Outgoing behavior based on his/her historical pattern.
	Monitor for excessive (Ex) transactions	Excessive ACH, ATM, Cash, Check, Monetary Instrument, POS and Wire activity. Also, excessive monetary instrument with same payee	- Detect Excessive Transaction Activity	- Identify an excessive frequency/amount of incoming or outgoing behavior or purchase behavior based on the peer group 90-day average. Identify groups that could be involved in money laundering, or smurfing through account relationships and the use of monetary instruments.
	Monitor for structuring activity	Structured Transaction Daily Scenario, Structured Transactions Monthly Scenario (2)	- Structuring	- Identify customers who are conducting multiple cash transactions (deposit or withdrawal) in increments under the reporting guidelines to avoid federal reporting. This could be a possible "red flag" for terrorist financing or a customer/member attempting to evade federal reporting requirements.
	Monitor for Keywords	ACH internet transfer list activity, cash deposits/ATM withdrawals in HR Country List, Cash with HR Volume POS Debit activity on a/c hitting human trafficking-smuggling list, HR country list for ACH, ATM and POS activity, and Large aggregate incoming/outgoing wires from/to HR country list and/or internet transfer lists	- Risky International Transfer	- Identify Human trafficking through flagging of travel and daily expenses, small dollar debit expenditures, ACH or POS activity.
	Monitor for high velocity	Velocity, Wire Velocity Activity (2)	- Velocity Aggregate - Velocity Single	- Identify deposits with depletion where the credits and debits to the account are roughly the same.
	Monitor for rapid paydown/ payoff of loans	Excessive ACH related to a loan, ACH Out of Pattern to a loan	- Overpayment of Loans Aggregated - Repayment of loan with Percentage of Loan Period	- Identify repayment of loan amount within a brief period after the loan was granted or when the payments are inconsistent with the original payment schedule.
Fraud including asset misappropriation and data theft.	Monitor for unusual returns, accounts opened with stolen identities and account takeovers	ACH Fraud, POS Fraud Returns, POS Fraud Chargeback (3)	- Inactive Account Fraud - Detect Excessive Transaction Activity - Dormant Account Aggregate - Dormant Account Single	- Identify Unusually large number of ACH returns, in comparison to other people in the population Identify accounts opened by fraudsters funded with compromised accounts where the rightful owners have noticed the attempt and cancelled the funding transfer, resulting in an ACH return. - Identify unusually large number of refund transactions, in comparison to other people in the population - Identify Unusually large number of chargebacks, in comparison to other people in the population - Detect accounts opened by fraudsters funded with compromised accounts where the rightful owners have noticed the attempt and cancelled the funding transfer, resulting in a chargeback.

AML Risk	Working Standard	Hummingbird System	Actimize System	Scenario
Cross-border movement of funds related to illicit activities	Monitor for Keywords	HR Country List ACH Activity, HR Country List ATM Activity, HR Country List POS Activity, ACH Internet Transfer List Activity, Cash Deposits/ATM Withdrawals in HR Country List, Large Aggregate Incoming or Outgoing Wires from or to HR Country List and/or Internet Transfer List (6)	- Risky International Transfer	- Identify large monthly volume of international ACH transactions, ATM transactions & POS Transactions involving countries deemed high risk. - Identify Large ACH transactions where the other party is a merchant that has been identified as having lax authentication practices - Identify domestic cash deposits that are followed by ATM withdrawals in high-risk countries. - Identify large aggregate incoming or outgoing wire transfers from or to HR countries.

System Rule Comparison

The transaction codes used for transaction monitoring in Actimize mirrored those used in HCM. Actimize rules includes the below transaction codes with specific threshold set for each different type of activity.

For ATM, Actimize defines that as a Channel of how a transaction is conducted instead of a transaction code.

Below are the transaction codes that are used in Actimize:	Below are the transaction codes that are used in HCM:
• ACH • Debit Card (POS) • Cash • Check • Check Cash (Cash) • Wire	• ACH • POS • ATM • Cash • Check • Wire

Parallel Testing Analysis

Parallel Scope 1 Cycle – October 1 to November 30, 2021:

Alerts Analysis

The below chart represents a comparison of alerts generated from October 1 to November 30, 2021, for transaction dates September 1 to October 31, 2021, for ABC/ABCX partner. During this cycle, the BSA Analytic team worked with Actimize's Professional Service team and Actimize engineers to setup the system and ensure the file intake protocol was adequate.

The large difference between Actimize alerts and HCM alerts is due to the initial implementation setup and required additional adjustment prior to Cycle 2 testing.

ABC/ABCX	Actimize Alert Volume	HCM Alerts Volume	Actimize-HCM Difference
Albert	0	0	0
Aspiration	5	116	-111
Aven	38	106	-68
BlueVine	146	1248	-1102
Brigit	0	0	0
Carta	0	8	-8
ABC	883	3130	-2247
Cheese	1	2	-1
Clearly	0	1	-1
Ellevest	0	11	-11
Fair	0	17	-17
Movo	0	12	-12
One Finance	162	1381	-1219
Possible	0	0	0
Sable	142	730	-588
Till	0	0	0
XI	0	0	0
TOTAL	**1377**	**6762**	**-5385**

A review of the Actimize alerts revealed 570 alerts (41%) also had HCM alerts and of those alerts, 57 were escalated to cases.

Case/SAR Analysis

As an additional adjustment to the data intake file protocol was necessary and comparable alert volume between Actimize and HCM was not achieved, the analysis of cases and SARs was not completed during this cycle.

Rule Adjustments

Actimize Professional Service team also identified that Burst in Beneficiary Activity rule was setup using external account instead of internal accounts. An adjustment to the Burst in Beneficiary Activity rule was approved and modified on November 11, 2021. It was also identified that the monthly rule, Activity Deviation, was setup using a 30-day rollup. This rule was changed to a 1-day rollup on October 31, 2021.

Parallel Scope 2 Cycle – December 2021:

Alerts Analysis

Below chart represents a comparison of alerts generated in December 2021, for transactions from November 1 to November 30, 2021, for ABC/ABCX partners. During this cycle, the BSA Analytic team worked with Actimize's Customer Success team and Actimize engineers to ensure the rules and system were working as expected.

The contrast between alerts generated in Actimize and HCM were decreased during this cycle. Actimize generated 30% more alerts than HCM during this cycle; however, it should be noted that ABC alerts represented 80% of the Actimize alert volume. ABC alert volume included many alerts for the same subject and once the alerts were reviewed by subject, decreased the volume by approximately 50%. The volume of alerts received across the ABCX partners between Actimize and HCM was lower than anticipated.

ABC/ABCX	Actimize Alert Volume	HCM Alerts Volume	Actimize-HCM Difference
Albert	1	0	1
Aspiration	78	81	-3
Aven	58	81	-23
BlueVine	106	991	-885
Brigit	0	0	0
Carta	0	13	-13
ABC	4,032	1,023	3,009
Cheese	2	5	-3
Clearly	0	2	-2
Ellevest	0	5	-5
Fair	1	20	-19
Movo	2	6	-4
One Finance	447	746	-299
Possible	0	0	0
Sable	283	525	-242
Till	0	0	0
XI	0	0	0
TOTAL	**5,010**	**3,498**	**1,512**

Case/SAR Analysis

An analysis of the cases created in Actimize revealed that 105 cases were created in Actimize as a result of HCM alerts generated in December 2021. Below chart summarizes the cases by type of suspicious activity and also identifies if an Actimize alert generated:

	Case Opened with HCM Alert	Volume of Cases with Actimize Alerts	Percent Difference
Cases - ALL	105	49	47%
Cases – AML	69	31	45%
Cases - Fraud	36	18	50%

All cases that were open from HCM alerts and were ML/TF related that did not have an Actimize alert were on Fintech partners. These cases represented only 36% of the total case population. Additionally, six cases, 5 AML cases and 1 Fraud case, were open from Actimize alerts that that did not alert in HCM. A SAR was filed on all but one of the AML cases.

Below chart represents the SARs filed from HCM alerts generated from December 1 to December 31, 2021, that didn't have an Actimize alert. Of the cases opened related to Cycle 2 alerts, we filed 20 SARs that were ML/TF related and Actimize didn't alert on the activity. These SARs were all on Fintech partners and were related to ACH, wires and debit card activity.

ABCX Partner	SARs Filed from HCM Alerts	ACH	Wires	Debit Card
BlueVine	6	4	2	
Cheese	1	1		
Fair	1	1		
Movo	1			1
OF	5	3		2
Sable	6	1	1	4
Grand Total	**20**	**10**	**3**	**7**

To ensure that Actimize alerts capture the above SAR activity, the rules were analyzed against the transactional activity to determine what adjustments were necessary.

Actimize Rule Adjustments

Pre-implementation tuning was conducted from January 10 to January 16, 2022, and based on the results, rule adjustments were recommended and approved on January 18, 2022. The adjustment to the system was finalized on January 21, 2022. These adjustments were necessary to address any Actimize rules that were not capturing activity that ABC BSA Department had filed an SAR on related to ML/TF activity. Two rules were found to be not performing as expected and were adjusted: (Note: for full details related to rule changes see Actimize Rules and Conditions – Recommendation – 1.21.22)

- Detect Excessive Transaction
  - Two ACH conditions were created
  - One Debit Card condition, the amount thresholds were adjusted
- Velocity Aggregate

- One wire condition was created
- One Debit Card condition, the day/amount thresholds were adjusted
- One ACH condition, the day/amount thresholds were adjusted
- One Debit Card condition, the day thresholds were adjusted
- Two ACH conditions, the day thresholds were adjusted

In addition to the tuning adjustment that occurred on 1/21/2022, it was also recommended and approved that the rollup setting be changed from 30 days to 7 days on all rules except the monthly rule, Activity Deviation, and the fraud rules. The primary reason for the change is to provide more visibility to the alert volume. Alerts will still be worked within a 30-day Service Level Agreement (SLA). Alerts that have additional incidents that occur after the alert generates, but before the alert has been triaged, will be added to the open alert via Actimize's modification functionality.

Below is Actimize's Reference Manual explanation of the Roll up period usage:

Roll up period:

Rollup period is a configurable parameter for a transaction monitoring rule.

If the rollup period is configured, an alert is shown to the analyst only at the end of the rollup period, counting from the date of detection of the first incident in the alert. There is no reset of the period when new incidents get added to the alert.

If the rollup period is not configured, an alert may be shown to a user as soon as it is generated (i.e., its first incident is detected). Every time that a new incident is rolled into the alert, this period is reset. The period during which a new incident can be added to the alert is determined by the configuration of the condition that generated that first incident. The period will have the same duration as what the condition uses to look for transactions. For example, if the condition monitors daily transactions, the period will be 1 day; If the condition monitors structuring behavior with a 14-day look-back window, the period will be 14 days (about 2 weeks).

Parallel Scope 3 Cycle:

Alerts Analysis

Below chart represents a comparison of alerts generated from January 1 to January 31, 2022, for ABC/ABCX partners. During this cycle, the BSA Analytic team continued to work with Actimize's Customer Success team and Actimize engineers to ensure the rules and system were working as expected.

Given the alert/case ratio in HCM was only 2.87% in January 2022, the variance between Actimize alerts and HCM is within acceptable ranges. The focus of Actimize tuning is to capture suspicious activity, while also reducing false positives.

ABC/ABCX	Actimize Alert Volume	HCM Alerts Volume	Actimize-HCM Difference
Albert	0	0	0
Aspiration	7	36	-29
Aven	56	96	-40
BlueVine	94	1147	-1053
Brigit	0	0	0
Carta	0	13	-13
ABC	2533	1777	756
Cheese	0	4	-4
Clearly	0	2	-2
Ellevest	1	7	-6
Fair	3	32	-29
Movo	4	11	-7
One Finance	166	1005	-839
Possible	0	0	0
Sable	192	741	-549
Till	0	41	-41
XI	0	0	0
TOTAL	**3056**	**4912**	**-1856**

Case/SAR Analysis

An analysis of the cases created in Actimize revealed that 185 cases were created in Actimize as a result of HCM alerts generated from January 1 to January 31, 2022.

All cases that were ML/TF related that did not have an Actimize alert were on the following Fintech partners: BlueVine, Cheese, Movo, One Finance and Sable. These cases only represented 17% of the total case population.

	Cases Opened with HCM Alerts	Volume of Cases with Actimize Alerts	Percent difference
Cases - ALL	185	94	51%
Cases – AML	81	49	60%
Cases - Fraud	104	46	44%

Of the 185 cases created, 81 cases were open due to ML/TF concern. Actimize alerts were identified on 49 of these cases, whereas 32 cases didn't have an Actimize alert.

An analysis of the 32 cases that didn't have an Actimize alert was conducted. This analysis included a review of the alerted activity in HCM and Actimize. The analysis found that within 28 of these cases that if the tuning adjustment that occurred on January 21, 202 had been in place an Actimize alert would have generated. The other 4 cases were found to be non-suspicious, and a SAR was not filed.

Additionally, 90 cases, 73 money laundering cases and 17 Fraud cases, were open from Actimize alerts that in which an HCM alert had not generated. As of this report, 73 cases have been decisioned and 41 SARs was filed. There were 30 money laundering related SARs and 11 SARs filed on fraudulent activity.

Rule Threshold Adjustments

As all SAR activity was either captured by Actimize rules or would have been captured based on the tuning that occurred on January 21, 2022, a rule adjustment for this cycle testing is not recommended.

Segmentation Analysis

ABC customers and accounts have a complex structure that then create dynamic alerts that spread across several customers and include transactions from many accounts. As a result of the complexity of the ABC alerts, segmentation rules are necessary to reduce false positives and increase the alert quality.

To assist with the calibration of the ABC Segmentation, SAS Analytic Software (SAS) was used to test possible thresholds. First Actimize rules were built in SAS. Then ABC customer transactional activity was run against Actimize's current rule thresholds for the rules in which ABC customers were primarily triggering:

- Detect Excessive
- Velocity Aggregate
- Velocity Single

Given that SAS and Actimize are different systems, the result variance was set at 90% match.

During January 2022, Actimize started to deploy Entity Group functionality. This functionality gives the ability to create customer segments based on same attributes and then apply that segment to rules for transaction monitoring.

This feature supports the following capabilities: -

1. Ability to create Entity Groups from the UI with the following combination:
    a. Entity Group Type -
        i. Criteria
    b. Segment Type -
        i. Account
        ii. Customer
        iii. Both
2. Ability to select Entity Groups in Transaction Monitoring rules.

ABC Customer Case/SAR Analysis

An analysis of the ABC customer cases created in Actimize revealed that 7 cases were created in Actimize as a result of HCM alerts generated from November 1,2021 to March 10, 2022. Actimize alerts generated on all ABC cases and therefore the implementation of the ABC segmentation doesn't impact the generation of these alerts.

Segmentation thresholds for ABC individuals and businesses that were tested in SAS were approved to be used in Actimize on February 14, 2022.

Actimize Benefits and Enhancement Recommendations

As part of the parallel testing, the BSA Analytic team and AML Investigation team were asked to provide feedback on the Actimize system. Below is a summary of the most common benefit/wish list the teams mentioned.

Benefits:

- Efficiency in clearing alerts
- Easier access to activity that triggered the alert
- Advance filter/sort functionality
- Quick link to customer static data

Enhancement Recommendations:

- SLA Warning Flag
- Customized Reports with more details about alerts/cases/SARs
- Disposition note field has a 1,000-character limit. Request this be expanded to 5,000.
- During filtering, dates do not stick between alerts. It requires the analysts to re-enter the filter date for each alert.

Enhancement requests have been submitted to Actimize and follow-up discussions will be conducted during the 2022/2023 Roadmap meeting.

Summary and Recommendation:

Parallel testing was conducted over three cycles: October-November 2021; December 2021 and January 2022. Over these cycles, adjustments were made to the Actimize transaction monitoring rules and settings to ensure that suspicious activity was being captured, while minimizing the false positives. The testing included comparing alerts in both systems and then ensuring Actimize alerts were generated on ML/TF related SARs filed from HCM alerts. During cycle two testing, adjustments to Actimize rules were recommended, approved, and implemented. Cycle three testing revealed that the adjustment made to the rules were sufficient to capture all ML/TF SAR activity and additional adjustment were not necessary. Additionally, during cycle two testing, the Roll-up setting was changed from 30 days to 7 days on most rules. This allowed for more visibility to the alert volume and improve workflow processing by displaying the alert earlier and then using Actimize's Modification Functionality to add additional incidents if they occur.

On February 23, 2022, Actimize deployed Entity Group functionality. This functionality had partially been deployed in January 2022 and then finalized in February 2022. As part of the Actimize transaction monitoring implementation, rule segmentation was also analyzed. Segmentation thresholds for ABC individuals and businesses had been tested using SAS. These thresholds were approved for use on February 14, 2022.

The overall performance of the Actimize alerts were compared to HCM alerts and used for the evaluation during this pre-implementation tuning exercise. During the last cycle of testing, it was confirmed that Actimize's transaction monitoring for ABC's and ABCX's customers captured all ML/TF SAR activity that was created from HCM alerts. Additionally, during cycle two and three, Actimize identified 47 customers that required SARs to be filed on their activity. Given these results it's recommended that Actimize be implemented as ABC Bank's transaction monitoring system of record effective March 1, 2022. It's additionally recommended that the alerts generating February 1, 2022, and after be consider production alerts.

Next Steps

A limited scope independent model validation has been scheduled to begin on March 14, 2022. This model validation will be conducted by AuditOne LLC and will include the following:

- Case Management – in production since 9/27/2021
- SAR filing – in production since 9/27/2021
- Watchlist Screening (OFAC and 314a) – in production since 2/1/2022
- Transaction Monitoring – parallel testing since 10/1 expected to be in production 3/1.

<u>Model Validation Tests</u>

Design Risk:

- Assess the completeness/quality of Management's document of the model.
- Verify that rules or scenarios are sufficient.

Input Risk:

- Assess the timeliness/effectiveness of the FI's internal data validation

- Assess the adequacy of change controls
- Assess the appropriateness of system access assignment
- Testing transactions

Performance Risk:

- Transaction testing
- Verify that alerts values are calculated accurately

Additionally, a full scope model validation will be conducted prior to the end of 2022.

All spreadsheets and images of internal and external documents/emails related to this tuning exercise are saved at the following location R:\BSA Department\6. Reports\Tuning Reports\2022-Q1 - Tuning Exercise\AML - Parallel Testing - Tuning